

Michigan Tech

Existence and Uniqueness for ODEs

Classical theorems and a discontinuous third-order equation

| Lecture 1 |

The textbook existence and uniqueness theorem

Theorem

Let $U \subset \mathbb{R} \times \mathbb{R}^n$ be open, with $(t_0, y_0) \in U$. Suppose $f : U \rightarrow \mathbb{R}^n$ and all partial derivatives

$$\frac{\partial f_i}{\partial y_j}, \quad 1 \leq i, j \leq n,$$

are continuous on U . Then, for some $h > 0$, the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0 \tag{1}$$

has a unique C^1 solution on $(t_0 - h, t_0 + h)$.

For a scalar equation, the hypotheses are continuity of f and $\partial f / \partial y$.

The conclusion gives both local existence and local uniqueness. Classical local theory:

The Picard–Lindelöf theorem

Theorem

Let $f : U \rightarrow \mathbb{R}^n$ be continuous on an open neighborhood U of (t_0, y_0) . Suppose there are $a, b > 0$ and $L \geq 0$ such that

$$Q = [t_0 - a, t_0 + a] \times \overline{B}(y_0, b) \subset U$$

and, for every $t \in [t_0 - a, t_0 + a]$ and $y, z \in \overline{B}(y_0, b)$,

$$\|f(t, y) - f(t, z)\| \leq L\|y - z\|. \quad (2)$$

Then the IVP has a unique local C^1 solution.

This is a local Lipschitz condition in y , uniform in t . Continuous state derivatives imply this condition on a sufficiently small cylinder. Teschl, Chapter 2: MIT

Carathéodory's existence theorem

Theorem

Let $a, b > 0$, $J = [t_0 - a, t_0 + a]$, and $Q = J \times \overline{B}(y_0, b)$. Suppose $f : Q \rightarrow \mathbb{R}^n$ satisfies:

1. For every fixed y , the function $t \mapsto f(t, y)$ is measurable.
2. For almost every t , the function $y \mapsto f(t, y)$ is continuous.
3. There is a nonnegative $m \in L^1(J)$ such that, for almost every t and every $y \in \overline{B}(y_0, b)$,

$$\|f(t, y)\| \leq m(t).$$

Then, for some $0 < h \leq a$, there is an absolutely continuous solution on $[t_0 - h, t_0 + h]$ with $y(t_0) = y_0$ and $y' = f(t, y)$ almost everywhere.

This theorem asserts existence. It does not assert uniqueness.

The solution in Carathéodory's theorem

An absolutely continuous solution satisfies the integral equation

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds. \quad (3)$$

Its derivative exists almost everywhere and satisfies the ODE there.

For example, a jump in *time* is allowed:

$$y' = H(t), \quad y(0) = 0, \quad y(t) = \begin{cases} 0, & t \leq 0, \\ t, & t > 0. \end{cases}$$

This solution has a corner at $t = 0$, but satisfies (??).

Continuity in the state variable remains a hypothesis. Carathéodory theory: Filippov, Chapter 1, section on continuity in the state and discontinuity in time.

Existence without uniqueness

Consider the scalar initial value problem

$$y' = 2\sqrt{|y|}, \quad y(0) = 0. \quad (4)$$

For every $c \geq 0$, define

$$y_c(t) = \begin{cases} 0, & t \leq c, \\ (t - c)^2, & t > c. \end{cases} \quad (5)$$

Every y_c solves (??). The function $y(t) \equiv 0$ also solves it.

The solution can remain at zero for any waiting time c before departing.

Continuity of f guarantees local existence, but allows nonuniqueness.

Checking the nonuniqueness example

For $t < c$, $y'_c = 0 = 2\sqrt{|y_c|}$. For $t > c$,

$$y'_c(t) = 2(t - c) = 2\sqrt{(t - c)^2} = 2\sqrt{|y_c(t)|}.$$

At $t = c$, both one-sided derivatives equal zero, so y_c is C^1 and the ODE holds there too.

For $y > 0$, the Lipschitz quotient at zero is

$$\frac{|f(y) - f(0)|}{|y - 0|} = \frac{2\sqrt{y}}{y} = \frac{2}{\sqrt{y}} \rightarrow \infty.$$

Thus f is continuous, but fails the local Lipschitz condition at zero.

Neither uniqueness theorem applies at the initial state.

Autonomous equations

Theorem

Let $V \subset \mathbb{R}^n$ be open, $y_0 \in V$, and $f : V \rightarrow \mathbb{R}^n$. For the initial value problem

$$y' = f(y), \quad y(t_0) = y_0, \quad (6)$$

the following conclusions hold:

1. If f is continuous, a local solution exists.
2. If f is locally Lipschitz, the local solution is unique.

In particular, $f \in C^1(V)$ guarantees local existence and uniqueness.

For existence, continuity gives a bound on a small closed ball, so Carathéodory's hypotheses hold. This continuous case is also Peano's theorem.

These are local conclusions. They do not guarantee existence for all time.

A state-dependent jump in a third-order equation

Consider the general model

$$y''' = g(y, y', y'') + \alpha H(y), \quad \alpha \neq 0, \quad (7)$$

where g is locally Lipschitz near the state of interest and

$$H(y) = \begin{cases} 0, & y < 0, \\ 1, & y > 0. \end{cases}$$

Choose any fixed finite value for $H(0)$. With $x = (y, v, a) = (y, y', y'')$, write

$$x' = f(x) = \begin{pmatrix} v \\ a \\ g(y, v, a) + \alpha H(y) \end{pmatrix}. \quad (8)$$

The jump occurs in the state variable y , across the surface $y = 0$.

Smooth branches and the solution concept

The two locally Lipschitz branch fields are

$$f_{-}(y, v, a) = \begin{pmatrix} v \\ a \\ g(y, v, a) \end{pmatrix}, \quad f_{+}(y, v, a) = \begin{pmatrix} v \\ a \\ g(y, v, a) + \alpha \end{pmatrix}.$$

Definition. A solution of (??) is an absolutely continuous state $x = (y, v, a)$ satisfying the equation almost everywhere.

Since $y' = v$ and $v' = a$, this gives $y \in C^2$. At an isolated crossing we match

$$y, \quad y', \quad y''$$

continuously, while y''' may jump.

Carathéodory's existence theorem does not directly apply to f at $y = 0$: state continuity fails.

Lemma: local existence and uniqueness at a crossing

Lemma (Transverse crossing)

Suppose g is locally Lipschitz on a neighborhood of $x_ = (0, v_*, a_*)$, where $v_* \neq 0$. Then there exists $h > 0$ such that (??), with*

$$x(t_*) = x_*,$$

has a unique local absolutely continuous solution on $(t_ - h, t_* + h)$. The solution crosses $y = 0$ once in this interval, with*

$$\operatorname{sgn} y(t) = \operatorname{sgn}(v_*(t - t_*)), \quad t \neq t_*. \quad (9)$$

The value assigned to $H(0)$ does not affect this solution in the almost-everywhere sense.

Proof: the initial velocity selects the branches

Every candidate solution satisfies $y' = v$ with continuous v . Hence

$$y(t) = v_*(t - t_*) + o(|t - t_*|). \quad (10)$$

Because $v_* \neq 0$, this proves (??) for sufficiently small $|t - t_*|$.

For $v_* > 0$, the branch selection is

	$t < t_*$	$t > t_*$
sign of $y(t)$	$y < 0$	$y > 0$
equation	$x' = f_-(x)$	$x' = f_+(x)$

For $v_* < 0$, the two branches exchange roles.

There is no interval of waiting on $y = 0$ when $v_* \neq 0$.

Proof: existence and uniqueness of the joined solution

Assume $v_* > 0$. Picard–Lindelöf gives unique branch solutions

$$x'_- = f_-(x_-), \quad x'_+ = f_+(x_+), \quad x_{\pm}(t_*) = x_*.$$

Each has first component satisfying (??). Therefore

$$x(t) = \begin{cases} x_-(t), & t \leq t_*, \\ x_+(t), & t \geq t_* \end{cases}$$

uses the correct branch on each side and is locally absolutely continuous. It solves (??) except possibly at t_* , which proves existence.

Every other solution has the same signs by (??). On each side it therefore solves the same branch IVP through x_* . Branch uniqueness proves uniqueness of the joined solution. The proof for $v_* < 0$ exchanges f_- and f_+ . □

The jump in the third derivative

At an upward crossing, $v_* > 0$,

$$\lim_{t \uparrow t_*} y'''(t) = g(0, v_*, a_*), \quad \lim_{t \downarrow t_*} y'''(t) = g(0, v_*, a_*) + \alpha.$$

Writing $\tau = t - t_*$, the one-sided expansions are

$$y(t_* + \tau) = v_*\tau + \frac{a_*}{2}\tau^2 + \begin{cases} \frac{g(0, v_*, a_*)}{6}\tau^3 + o(|\tau|^3), & \tau < 0, \\ \frac{g(0, v_*, a_*) + \alpha}{6}\tau^3 + o(|\tau|^3), & \tau > 0. \end{cases}$$

The value, slope, and second derivative agree at the join.

For a downward crossing the temporal jump in y''' is $-\alpha$.

Tangency and the limits of the crossing lemma

When $v_* = 0$, the transverse crossing lemma does not apply. If $a_* \neq 0$, every candidate solution satisfies

$$y(t_* + \tau) = \frac{a_*}{2}\tau^2 + o(\tau^2).$$

Thus y has the same sign on both sides of t_* : the trajectory touches the surface locally.

If $v_* = a_* = 0$, this expansion does not determine the side. Further analysis of the equation and the solution concept is necessary.

Failure of a theorem's hypotheses does not by itself prove nonuniqueness.

Filippov's theory provides a broader framework for discontinuous vector fields. The transverse case above follows directly from classical branch uniqueness.

References

Classical existence and uniqueness

G. Teschl, *Ordinary Differential Equations and Dynamical Systems*, Graduate Studies in Mathematics 140, AMS, 2012, Chapter 2.

[Author's book page and freely available text](#)

MIT OpenCourseWare, *18.100B Real Analysis*, Spring 2025, Lecture 23: Existence & Uniqueness for ODEs: Picard–Lindelöf Theorem.

[Lecture notes](#)

Carathéodory theory and discontinuous equations

A. F. Filippov, *Differential Equations with Discontinuous Righthand Sides*, edited by F. M. Arscott, Kluwer, 1988. Chapter 1 includes equations continuous in the state and discontinuous in time.

[doi:10.1007/978-94-015-7793-9](https://doi.org/10.1007/978-94-015-7793-9)

The transverse crossing lemma here is an elementary consequence of Picard–Lindelöf, with its proof included in the deck.